Theoretical and Applied Climatology (2026) 157:606
https://doi.org/10.1007/s00704-026-06536-w

RESEARCH

Skill assessment of AI-based weather model in heavy precipitation prediction over South Korea

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Received: 12 February 2026 / Accepted: 14 August 2026

Abstract

As AI-based weather prediction model emerges, its prediction skill for extreme weather draws great attention. This study assesses the skill of GraphCast in heavy precipitation prediction against rain gauge observations focusing on 10 heavy precipitation cases in South Korea. For comparisons, the Weather Research and Forecasting (WRF) model simulations with different horizontal resolutions of 27 (WRF_{27km}), 9 (WRF_{9km}), and 3 km (WRF_{3km}) are used. When the skill in predicting general patterns of precipitation is evaluated using the normalized mean bias, normalized centered root-mean-square error, and spatial correlation coefficient, GraphCast shows better or comparable skill compared with WRF_{27km}, WRF_{9km}, and WRF_{3km}. When the skill for different precipitation thresholds is separately examined using the equitable threat score (ETS), neighborhood ETS, and other contingency table-based metrics, GraphCast shows relatively large overprediction of light precipitation (≤20 mm) and underprediction of heavy precipitation (≥70 mm) compared with WRF_{27km} as well as WRF_{9km} and WRF_{3km}. GraphCast reproduces observed topographic enhancements of precipitation over the Sobaek Mountains by 42%, which is less than those reproduced by WRF_{27km} (71%), WRF_{9km} (63%), and WRF_{3km} (90%). While the synoptic conditions and their associated water vapor transports are similarly simulated by GraphCast and WRF_{27km}, the upward motions over the Sobaek Mountains are weaker for GraphCast than for WRF_{27km}. The underpredictions of heavy precipitation and topographically enhanced precipitation by GraphCast are mainly associated with its weak ability to predict large precipitation peaks with scales of about 25–100 km.

Keywords AI-based weather prediction model · GraphCast · Heavy precipitation · Topographic effect · South Korea

1 Introduction

Prediction of extreme precipitation events is important in weather forecasting (Majumdar et al. 2021; Busker et al. 2025), since extreme precipitation events result in substantial loss of life and damage to property. For example, in South Korea, record-breaking heavy precipitation events occurred in summer 2020 and caused 46 fatalities and missing persons and the property damage of ~860 million dollars (KMA 2021). In China, torrential rainfall occurred in Henan Province in July 2021 and resulted in 398 deaths and missing persons and the direct economic losses of ~19 billion dollars (Zhang et al. 2023). As the occurrence frequency of extreme precipitation events is expected to increase under climate change (IPCC 2021), accurate prediction of extreme precipitation events becomes increasingly important.

In recent years, several technology companies and/or researchers developed different data-driven artificial intelligence (AI)-based weather prediction (AIWP) models such as the Fourier Forecasting Neural Network (FourCastNet) from NVIDIA (Pathak et al. 2022), Pangu-Weather from Huawei (Bi et al. 2023), GraphCast from Google (Lam et al. 2023), and the AI Forecasting System (AIFS) from the European Centre for Medium-Range Weather Forecasts (ECMWF) (Lang et al. 2024). As these models have shown good performance as well as very fast computation speeds (Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023; Lang et al. 2024), AIWP model has emerged as a promising new

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Published online: 08 September 2026

weather prediction system. This new type of model predicts a future atmospheric state based on historical evolutions of atmospheric state in past data, which is different from traditional numerical weather prediction (NWP) model that predicts based on physical governing equations. To further develop and improve AIWP model for introducing it into operational forecasting and/or weather research in earnest, extensive examinations of strengths and weaknesses of AIWP model are essential (Ben Bouallègue et al. 2024; Radford et al. 2025b).

Early studies evaluating the AIWP models showed that compared with the Integrated Forecasting System (IFS) model of the ECMWF, the AIWP models show competitive or even better scores for many meteorological variables including geopotential and temperature in terms of primary metrics such as the root-mean-square error (RMSE) and/or anomaly correlation coefficient (Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023; Lang et al. 2024). Furthermore, subsequent studies reported that the skill of the AIWP models is comparable to or better than the skill of IFS model in also predicting hot or cold extremes (Ben Bouallègue et al. 2024; Olivetti and Messori 2024; Pasche et al. 2025), tropical cyclone tracks (Ben Bouallègue et al. 2024; Liu et al. 2024; DeMaria et al. 2025), and overall characteristics of strong extratropical cyclones (Magnusson 2023; Charlton-Perez et al. 2024). These results suggest that the current versions of AIWP models may already surpass NWP model for predicting synoptic- and large-scale evolutions of pressure and temperature and are also skillful at predicting several types of extreme weather events.

In addition to pressure and temperature, precipitation amount is a crucial predictand in weather forecasting. Precipitation amount is spatially heterogeneous (e.g., Donat et al. 2014) and exhibits highly skewed distributions (e.g., Casati et al. 2008), thereby necessitating evaluations from different perspectives using various metrics. Several studies evaluated precipitation prediction skill of the AIWP models. Lam et al. (2023) showed that for 6-h accumulated precipitation amounts, GraphCast exhibits better scores than IFS in terms of the RMSE and stable equitable error in probability space (SEEPS). Lang et al. (2024) reported that AIFS exhibits better scores of SEEPS than IFS at longer lead times for 24-h accumulated precipitation amounts. Through the evaluation of GraphCast over China during 2020–2021, Yan et al. (2025) showed that for 1–3 day precipitation forecasts, GraphCast tends to exhibit greater skill than IFS in terms of correlation coefficient, mean error, and probability of detection. Radford et al. (2025a) compared precipitation forecasts of GraphCast initialized by the IFS and Global Forecast System (GFS) analyses with the forecasts of IFS and GFS over the USA from 1 January 2022 to 1 October 2023. They found that while GraphCast shows better scores of RMSE and SEEPS than IFS and GFS, IFS and GFS better reproduce the distribution of precipitation intensity and show greater gains in fractional skill score with increasing neighborhood size than GraphCast. By conducting a comparative analysis of precipitation prediction skill of AIFS and IFS over China from April 2024 to March 2025, Pan et al. (2025) revealed that AIFS shows better scores of RMSE and mean error than IFS but IFS better captures precipitation variabilities.

While overall performance of AIWP model for precipitation prediction was investigated by the previous studies, its skill in predicting heavy/extreme precipitation events is relatively less examined. In-depth understanding of prediction skill of AIWP model in heavy/extreme precipitation events is vital for its use in high-impact weather forecasting. Another important aspect is the physical consistency of simulations from AIWP model. As the AIWP models are not constrained to satisfy physical governing equations in contrast to NWP models, simulation results of the AIWP models are likely to exhibit a higher degree of physical inconsistency than those of NWP models. While the physical consistency of winds simulated by AIWP model was examined in terms of the geostrophic wind balance and the relative magnitude of rotational and divergent wind components (Bonavita 2024), the physical consistency of precipitation simulated by AIWP model has not been examined yet to the authors' knowledge.

In this study, we assess precipitation prediction skill of GraphCast focusing on heavy precipitation cases in the summer of 2020 in South Korea that caused significant casualties and economic losses (KMA 2021). Examinations of these cases can help to better understand the skill of current AIWP model for heavy/extreme precipitation. Also, as one aspect of physical consistency, how much topographic enhancements of precipitation are reproduced by GraphCast for these precipitation cases is evaluated. For comparisons with GraphCast simulations, simulations from the Weather Research and Forecasting (WRF) model are used. The WRF model is chosen for conducting numerical weather simulations with various resolutions. Since numerical simulations with different resolutions are expected to have different degrees of prediction skill in heavy/extreme precipitation, they can help to obtain in-depth understanding of AIWP skill through comparisons.

2 Data and methods

2.1 Observation data and simulation settings

The 10 days exhibiting the largest accumulated precipitation amounts over South Korea among summer days in 2020 are chosen as the heavy precipitation cases of interest which are 29 June; 10, 13, 23, and 29 July; 6, 7, 8, 10, and 11 August (Jin and Baik 2025). These days feature the daily maximum precipitation amounts exceeding 130 mm and the daily precipitation amounts averaged over South Korea ranging from ~30 to ~70 mm (Jin and Baik 2025). The cases are numbered from 1 to 10 in chronological order. While rain gauge data are widely regarded as the most reliable precipitation observation data (e.g., Chen et al. 2017; Meyer et al. 2017), evaluations of precipitation simulated by AIWP models were mainly conducted using precipitation reanalysis and/or analysis data (e.g., Pathak et al. 2022; Lam et al. 2023; Lang et al. 2024; Radford et al. 2025a). In this study, 24-h accumulated precipitation amounts during 0000–2400 LST on the 10 days observed at 549 rain gauges in South Korea are used for evaluation. This rain gauge network is comprised of automated synoptic observing system (ASOS) and automatic weather system (AWS) stations which are operated by the Korea Meteorological Administration (KMA) (https://data.kma.go.kr).

GraphCast (GC) is an AIWP model developed by Google, which autoregressively predicts various meteorological variables including precipitation with a 6-h time interval (Lam et al. 2023). Its architecture is based on graph neural networks in an encoder-processor-decoder configuration with icosahedral meshes of different resolutions. The version of GC used in this study has a horizontal resolution of 0.25° (~25 km in and around South Korea) and 37 pressure levels. This version of GC is trained on the ECMWF Reanalysis version 5 (ERA5) data (Hersbach et al. 2020) from 1979 to 2017. The GC simulation for each heavy precipitation day is initialized at 0000 LST on the day of interest using the ERA5 data. The 24-h accumulated precipitation amounts calculated from 6-h accumulated precipitation outputs at 0600, 1200, 1800, and 2400 LST are used for analysis after bilinearly interpolated to the locations of rain gauges.

The WRF model is an NWP model designed for both atmospheric research and forecasting applications (Skamarock et al. 2019). The version 4.3.3 of WRF model is used in this study. Three one-way nested domains (Fig. 1a) with horizontal resolutions of 27, 9, and 3 km and 44 vertical layers are used and the ERA5 data are used as initial and boundary conditions for the WRF simulations. For each heavy precipitation day, 27 combinations of WRF simulations are performed by adopting 3 microphysics schemes, 3 boundary-layer schemes, and 3 radiation schemes which are summarized in Table 1 together with other employed physics parameterization schemes. The WRF simulations for each heavy precipitation day are initialized at 1200 LST on the previous day and integrated up to 2400 LST on the day of interest. The precipitation amounts accumulated during the last 24 h averaged over the 27 simulations are bilinearly interpolated to the locations of rain gauges and used for analysis. To compare precipitation simulated by WRF with different horizontal resolutions, the simulations in the domains 1, 2, and 3 are named WRF_{27km}, WRF_{9km}, and WRF_{3km}, respectively, and are separately examined. Note that the horizontal resolutions of GC and WRF_{27km} are similar to each other and the horizontal resolution of WRF_{9km} is the same as the resolution of IFS high-resolution forecast system (HRES) that was frequently compared with AIWP models in previous studies (e.g., Lam et al. 2023; Ben Bouallègue et al. 2024; Lang et al. 2024; Olivetti and Messori 2024).

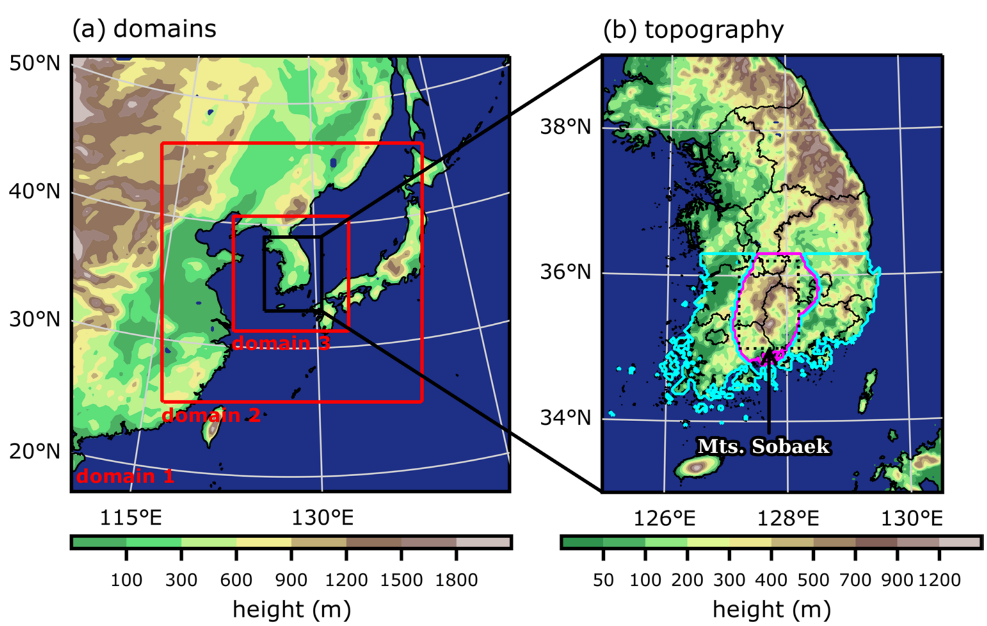

Fig. 1 (**a**) Three one-way nested domains used for the WRF simulations and (**b**) topography of South Korea. In Fig. 1b, the pink contour indicates the TP region and the sky-blue contour indicates the NT region

Table 1 Physics parameterization schemes used for the WRF simulations

Physics	Parameterization scheme
Microphysics	Morrison double-moment scheme (Morrison et al. 2009) WRF double-moment 6-class scheme (Lim and Hong 2010) Thompson aerosol-aware scheme (Thompson and Eidhammer 2014)
Boundary-layer/ surface-layer physics	Yonsei University scheme (Hong et al. 2006)/ revised MM5 scheme (Jiménez et al. 2012) Mellor–Yamada–Nakanishi–Niino (MYNN) Level 2.5 scheme (Nakanishi and Niino 2006)/ MYNN scheme (Olson et al. 2021) University of Washington scheme (Bretherton and Park 2009)/revised MM5 scheme
Radiation physics	Community Atmosphere Model scheme (Collins et al. 2004) Rapid Radiative Transfer Model for general circulation models scheme (Iacono et al. 2008) New Goddard scheme (Chou and Suarez 1999; Chou et al. 2001)
Cumulus physics	Kain–Fritsch scheme (Kain 2004) (only for the domains 1 and 2)
Land surface physics	Unified Noah land surface model (Tewari et al. 2004)

2.2 Evaluation metrics

Various metrics are computed for assessing precipitation prediction skill. To evaluate overall patterns of simulated precipitation, the normalized mean bias (NMB), normalized centered root-mean-square error (NCRMS), and spatial correlation coefficient (SCC) for each heavy precipitation case are computed as follows:

$$\mathrm{NMB} = \frac{\bar{f} - \bar{o}}{\bar{o}} \times 100, \tag{1}$$

$$\mathrm{NCRMS} = \frac{\sqrt{\frac{1}{M}\sum_{m}^{M}\left\{(f_m - o_m) - (\bar{f} - \bar{o})\right\}^2}}{\bar{o}} \times 100, \tag{2}$$

$$\mathrm{SCC} = \frac{\sum_{m}^{M}(f_m - \bar{f})(o_m - \bar{o})}{\sqrt{\sum_{m}^{M}(f_m - \bar{f})^2}\sqrt{\sum_{m}^{M}(o_m - \bar{o})^2}}, \tag{3}$$

where M is the total number of rain gauges (i.e., 549), m is the rain gauge index, f_m and o_m are the simulated and observed 24-h accumulated precipitation amounts at the location of the m-th rain gauge, respectively, and $\bar{f} = \sum_{m=1}^{M} f_m \big/ M$ and $\bar{o} = \sum_{m=1}^{M} o_m \big/ M$ are the simulated and observed 24-h accumulated precipitation amounts averaged over the locations of all rain gauges, respectively. The normalized metrics (i.e., NMB and NCRMS) are chosen instead of the simple mean bias and RMSE, to avoid overweighting the errors from heavy precipitation cases having relatively large precipitation amounts. Note that the NCRMS and SCC do not take into account the errors explained by mean bias.

Table 2 2×2 contingency table for a given precipitation threshold

		Event observed?		
		Yes	No	
Event forecast?	Yes	a	b	$a+b$
	No	c	d	$c+d$
		$a+c$	$b+d$	

To evaluate simulated precipitation separately for different magnitudes, a contingency table (Table 2) is constructed for different precipitation thresholds by considering all 10 precipitation cases. Based on the contingency table, the equitable threat score (ETS) (Schaefer 1990) is computed:

$$\mathrm{ETS} = \frac{a - a_{\mathrm{random}}}{a + b + c - a_{\mathrm{random}}}, \tag{4}$$

where a is the number of hits, b is the number of false alarms, c is the number of misses, d is the number of correct negatives, and a_{random} is the expected number of hits purely due to random chance calculated as

$$a_{\mathrm{random}} = \frac{(a+b)(a+c)}{a+b+c+d}. \tag{5}$$

In addition, performance diagram is plotted, which allows a collective examination of the success ratio (SR), probability of detection (POD), frequency bias (FB), and threat score (TS) that are defined as follows:

$$\mathrm{SR} = \frac{a}{a+b}, \tag{6}$$

$$\mathrm{POD} = \frac{a}{a+c}, \tag{7}$$

$$\mathrm{FB} = \frac{a+b}{a+c}, \tag{8}$$

$$\mathrm{TS} = \frac{a}{a+b+c}. \tag{9}$$

Even though magnitudes of precipitation peaks are reasonably predicted, prediction skill can be undervalued by the above metrics due to positional errors of simulated precipitation peaks. To examine this aspect, the neighborhood

ETS (nETS) defined by Clark et al. (2010) is computed for different radii. The nETS is calculated from a contingency table that has a more relaxed criterion of hits by considering a neighborhood defined by a given radius. Detailed information of nETS is given in Clark et al. (2010) and Schwartz (2017).

3 Results and discussion

3.1 General precipitation patterns

The spatial distributions of observed and simulated 24-h accumulated precipitation amounts for the 10 heavy precipitation

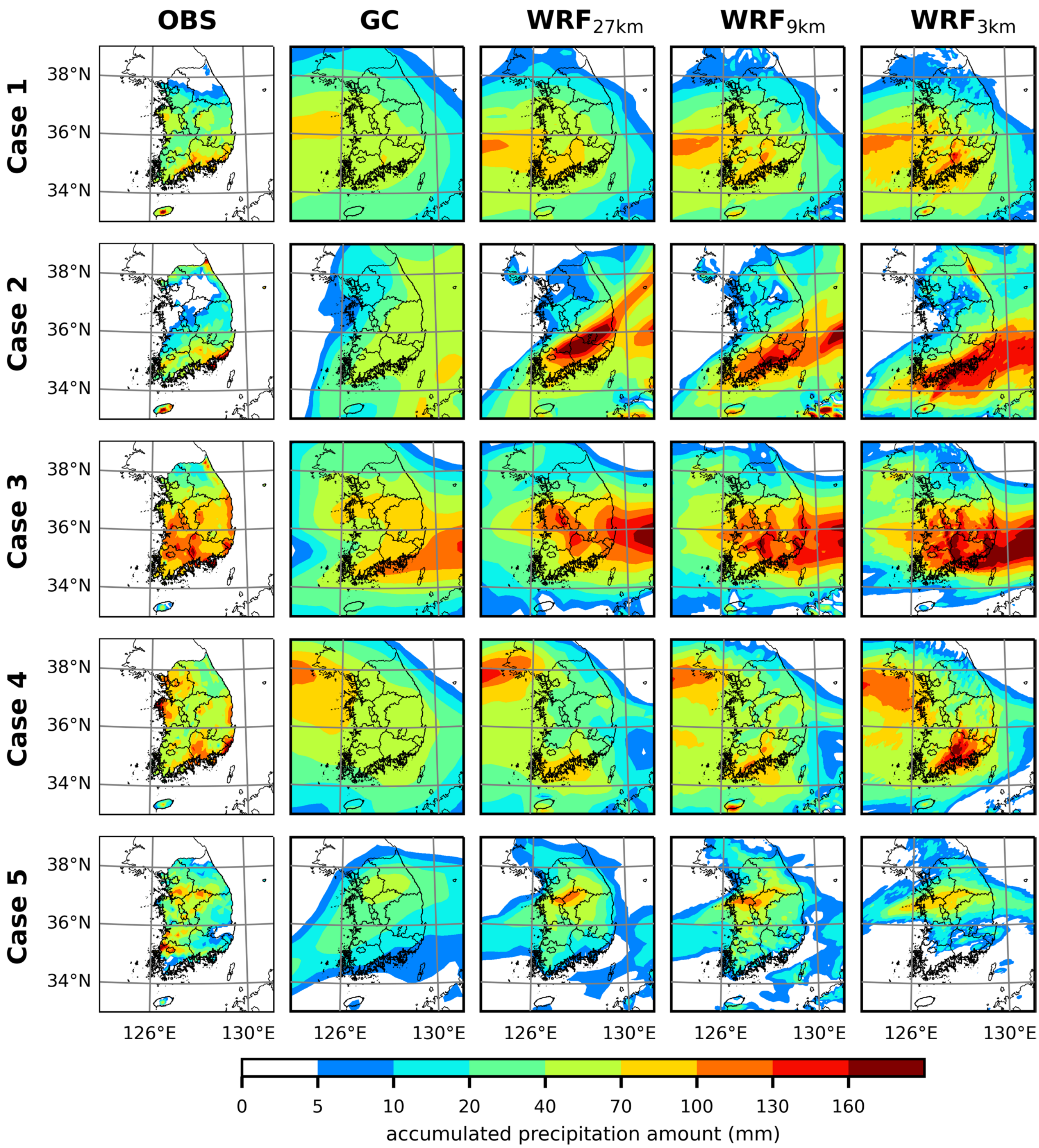

Fig. 2 Spatial distributions of 24-h accumulated precipitation amount in the (first column) rain gauge observation and in the (second column) GraphCast, (third column) WRF$_{27km}$, (fourth column) WRF$_{9km}$, and (fifth column) WRF$_{3km}$ simulations for Cases 1–5

cases are presented in Figs. 2 and 3. GC well captures the general patterns of observed precipitation, despite its small computational burdens. For example, for Case 1, GC predicts relatively large precipitation west of South Korea and in the southern part of South Korea, which resembles the relatively large precipitation observed in the western part and southern part of South Korea. Also, for Case 8, GC predicts a band of large accumulated precipitation over the southeastern part of South Korea, which is similar to the large precipitation observed over the southern part of South Korea. Meanwhile, GC produces smoothed precipitation patterns across the 10 cases, which is partly associated with its relatively coarse resolution of 0.25°. GC tends to predict wider areas of appreciable precipitation (>5 mm) compared with the observed areas but

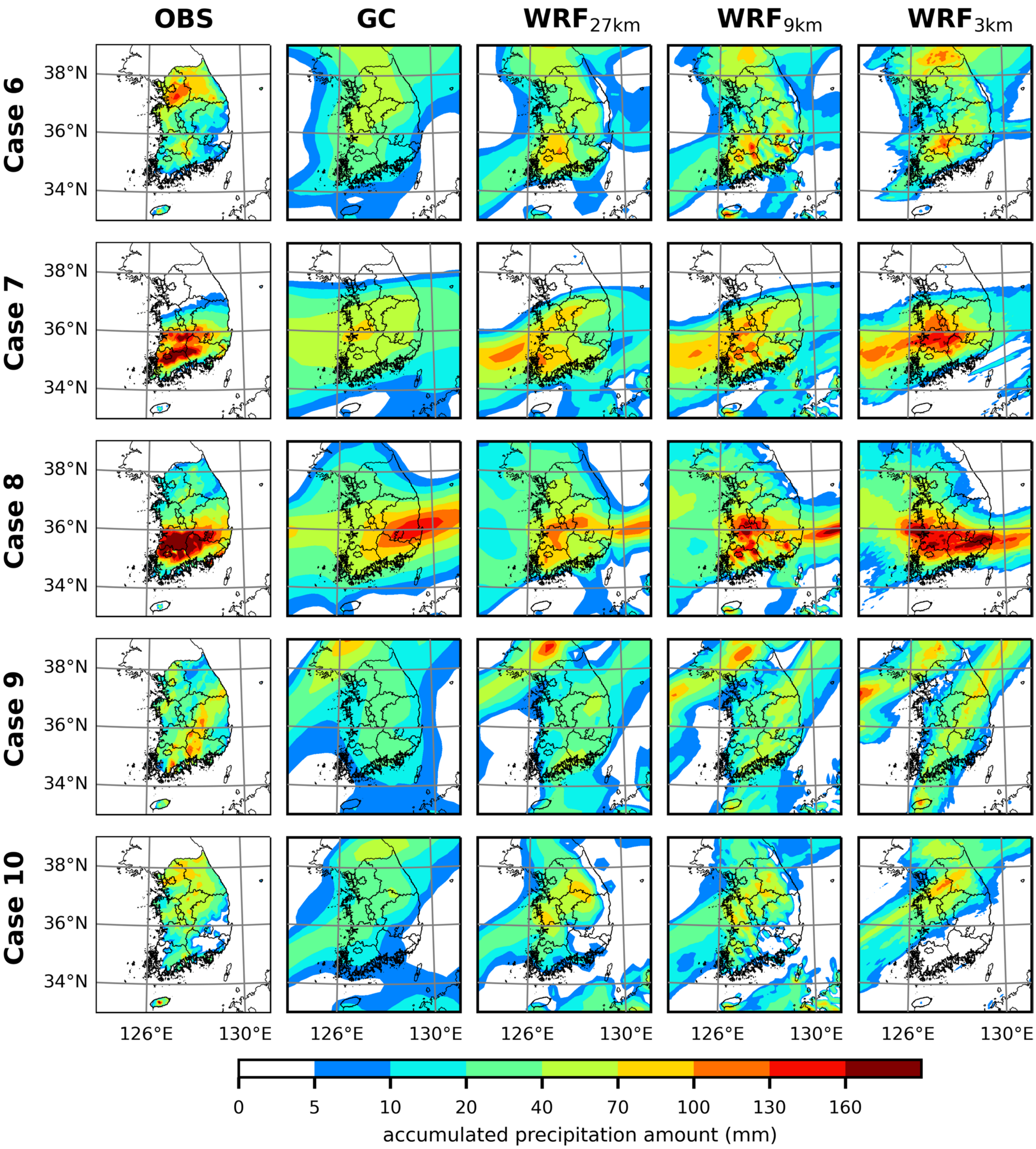

Fig. 3 As in Fig. 2, but for Cases 6–10

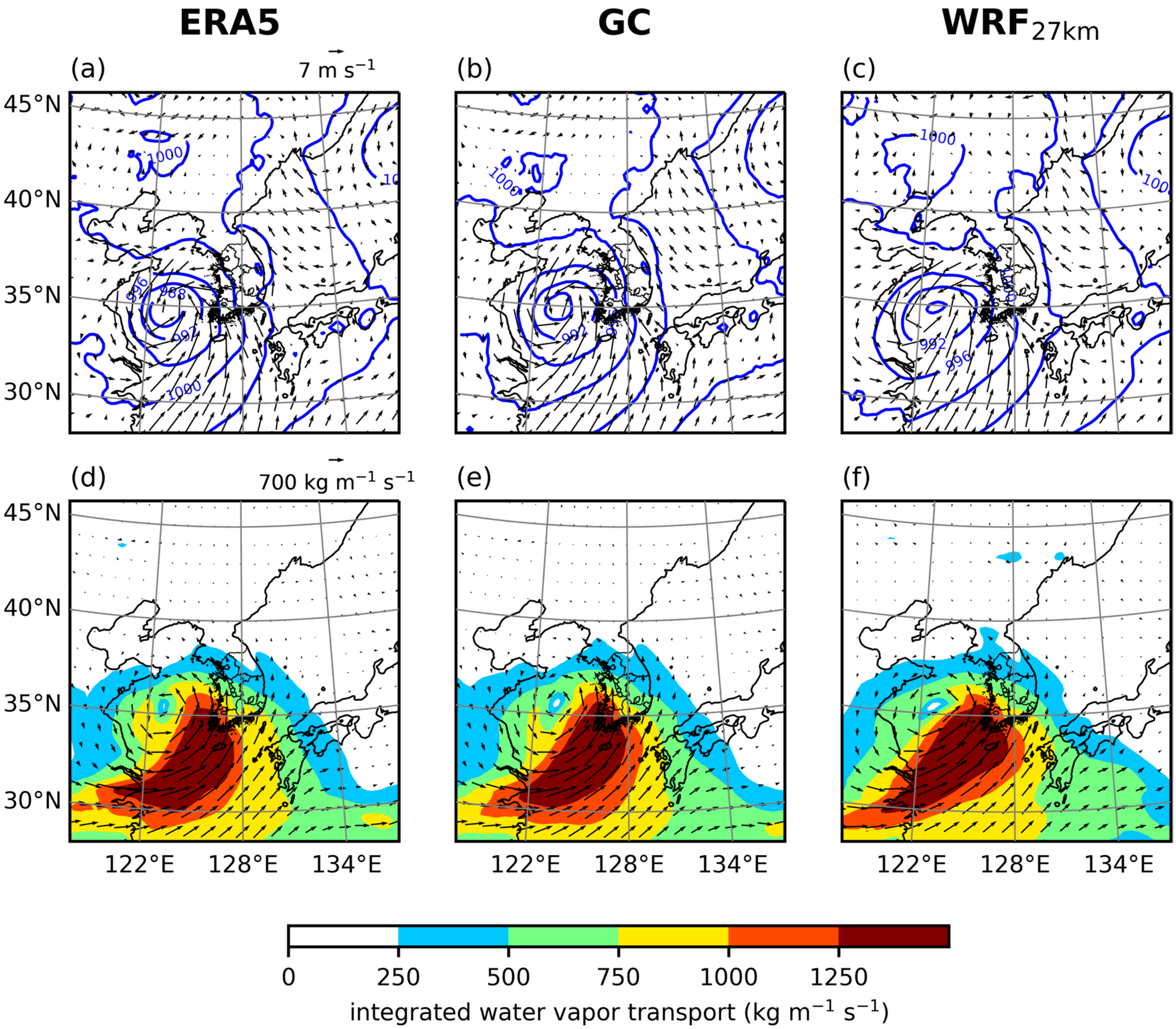

Fig. 4 Fields of (first row) sea-level pressure (blue contours), 10-m wind vector (arrows), and (second row) integrated water vapor transport (shades and arrows) at 1800 LST for Case 1 in the (first column) ERA5 data and (second column) GraphCast and (third column) WRF_{27km} simulations

misses the observed precipitation peaks higher than 100 mm in South Korea for 8 out of the 10 cases (i.e., all cases except for Cases 3 and 8). WRF_{27km} also produces relatively smoothed precipitation patterns, similar to GC, but misses the observed peaks higher than 100 mm in South Korea for 4 out of the 10 cases (i.e., Cases 1, 4, 9, and 10). WRF_{9km} and WRF_{3km} tend to capture precipitation patterns with scales finer than ~100 km and better reproduce the magnitudes of observed peaks in many cases, although the locations of simulated peaks often deviate from the locations of observed peaks.

The skill of GC in predicting synoptic conditions and their associated moisture transports in the heavy precipitation cases is exemplified in Figs. 4 and 5, which present the fields of sea-level pressure, 10-m wind vector, and integrated water vapor transport for Case 1 and Case 8 in the ERA5 data and GC and WRF_{27km} simulations. The integrated water vapor transport is calculated for the 1000–300-hPa layer. For Case 1, a strong low-pressure system with its central pressure reaching 988 hPa approaches South Korea from the west (Fig. 4a). GC simulates the pressure and wind patterns that are very similar to those in the ERA5 data (Fig. 4b). WRF_{27km} also adequately reproduces the strong low-pressure system but the similarity in pressure and wind patterns is relatively low compared with GC (Fig. 4c). The southwesterlies associated with the strong low-pressure system cause very strong moisture transport well exceeding 1250 kg m^{-1} s^{-1} toward South Korea (Fig. 4d). Both GC and WRF_{27km} well capture this strong moisture transport accompanying the low-pressure system (Fig. 4e and f).

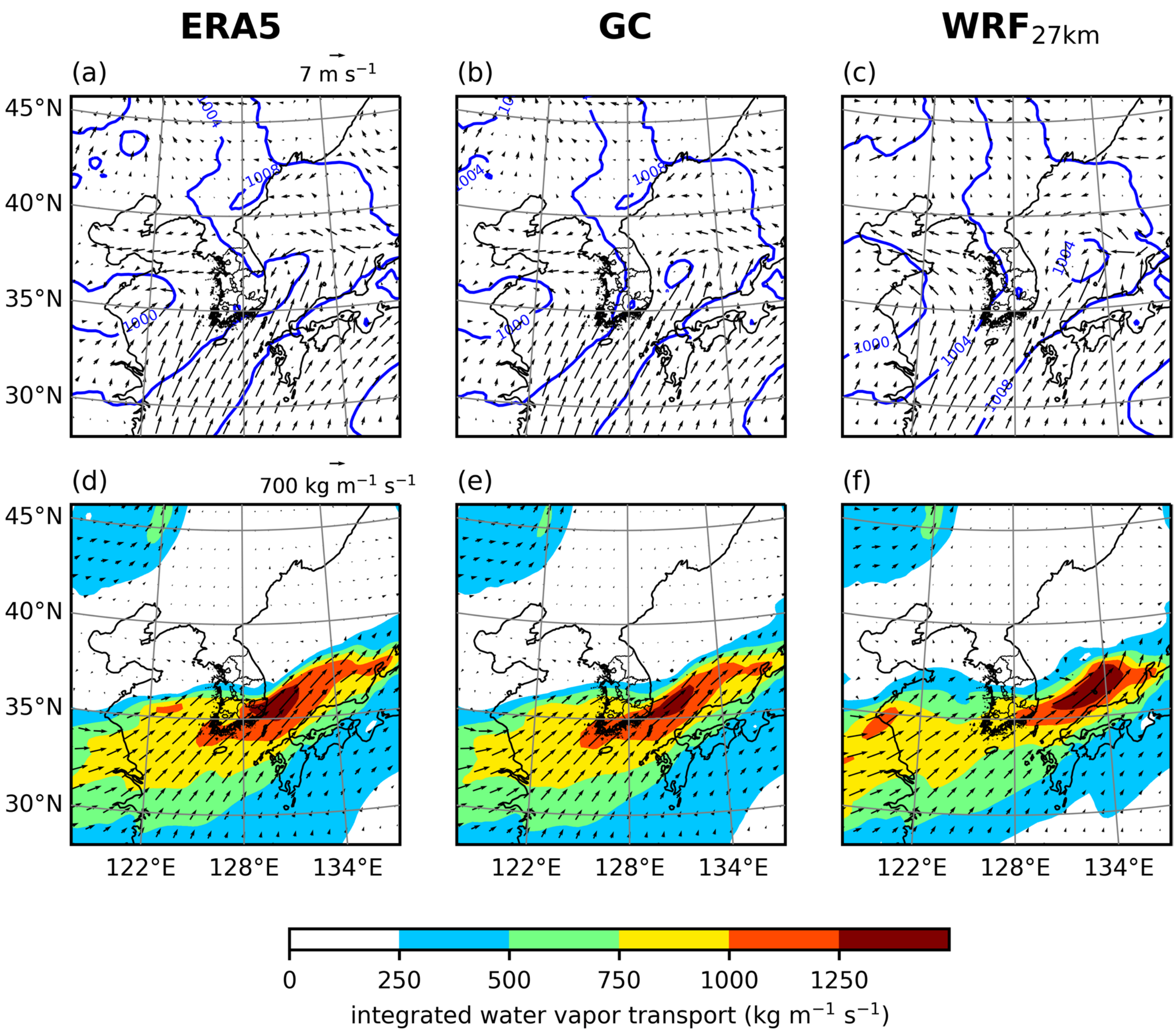

Fig. 5 As in Fig. 4, but at 0600 LST for Case 8

For Case 8, the pressure is relatively low west of South Korea and relatively high southeast of South Korea, exhibiting a northwest–southeast gradient (Fig. 5a). This pressure distribution induces prevalent southwesterlies south of and east of South Korea. This is associated with the quasi-stationary monsoon front of the East Asian summer monsoon. GC and WRF_{27km} capture the pressure distribution and resultant southwesterlies (Fig. 5b and c). The prevalent southwesterlies lead to a band of strong moisture transport, supplying abundant moisture to the southern part of South Korea (Fig. 5d). GC better reproduces the overall pattern and peak location of moisture transport than WRF_{27km}, though both models adequately simulate the moisture transport band (Fig. 5e and f). The close agreement between the ERA5 data and GC prediction presented in Figs. 4 and 5 indicates a very good performance of GC for predicting synoptic conditions and their associated moisture transports in heavy/extreme precipitation events.

Figure 6 shows the medians and interquartile ranges of NMBs, NCRMSs, and SCCs of Cases 1–10 for the GC, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations. All models tend to exhibit negative NMBs over South Korea for the heavy precipitation cases (Fig. 6a). GC exhibits negative NMBs of ~14% whose magnitudes are similar to those found in WRF_{27km} (~13%). When the horizontal resolution of WRF increases to 9 km or higher, the magnitudes of negative NMBs in WRF are reduced. For the NCRMS, GC shows the best performance among the models (Fig. 6b). The WRFs show similar magnitudes of NCRMSs that are slightly larger than the NCRMSs of GC, with WRF_{3km}

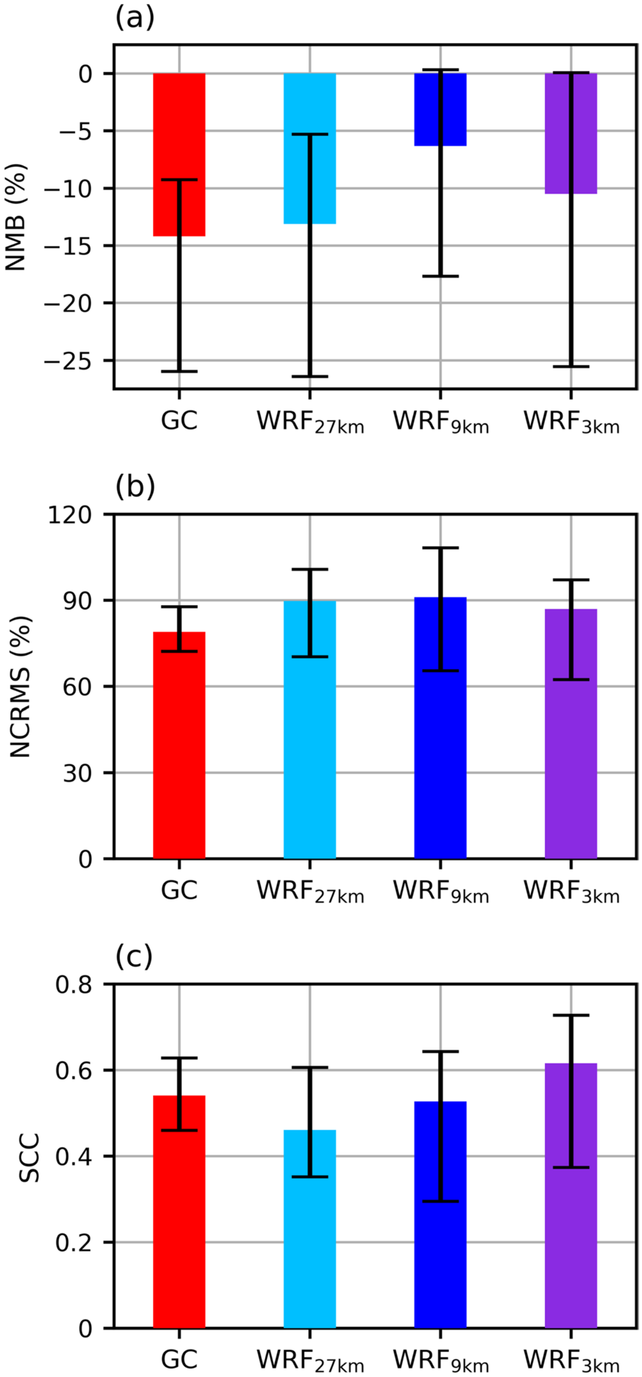

Fig. 6 Medians (bars) and interquartile ranges (whiskers) of (**a**) normalized mean biases, (**b**) normalized centered root-mean-square errors, and (**c**) spatial correlation coefficients of Cases 1–10 for the GraphCast, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations

exhibiting relatively small NCRMSs among the WRFs. GC has SCCs of ~0.54 which are higher than the SCCs of WRF_{27km} (~0.46) (Fig. 6c). The SCCs of WRF increase with increasing horizontal resolution, with the SCCs of WRF_{3km} being ~0.62 that are higher than the SCCs of GC. Overall, in terms of the metrics evaluating general patterns of precipitation, GC shows a better or comparable performance compared with the WRFs having similar (WRF_{27km}) and higher resolutions (WRF_{9km} and WRF_{3km}). The excellent skill of GC in predicting general patterns of precipitation in heavy precipitation events is qualitatively in line with the evaluation results for other meteorological variables (e.g., geopotential and temperature) in previous studies reporting better or competitive performances of AIWP models compared with NWP models in terms of the RMSE and/or anomaly correlation coefficient (e.g., Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023; Lang et al. 2024).

3.2 Heavy precipitation

To look into prediction skill for different precipitation magnitudes, the ETSs with precipitation thresholds ranging from 5 to 100 mm for the GC, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations are examined in Fig. 7. GC shows relatively low ETS at light precipitation (≤10 mm) and heavy precipitation (≥70 mm) thresholds and exhibits relatively high ETS at moderate precipitation thresholds (30–60 mm). WRF_{27km} shows a decreasing tendency in ETS with increasing precipitation threshold, exhibiting better scores at the light precipitation thresholds, worse scores at the moderate precipitation thresholds, and similar or slightly better scores at the heavy precipitation thresholds compared with GC. WRF_{9km} and WRF_{3km} show higher ETSs than WRF_{27km} at most of the precipitation thresholds except for the light precipitation thresholds. For WRF_{3km}, its ETS is even higher than the ETS of GC at all precipitation thresholds. The ETS of GC is considerably lower than the ETSs of WRF_{9km} and WRF_{3km} at the heavy precipitation thresholds.

Figure 8 shows the nETSs as functions of radius with precipitation thresholds of 10, 40, 70, and 100 mm for the GC, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations. The nETS increases with increasing radius for all models, since a more relaxed criterion of hits leads to higher scores. For all of the four thresholds, the increase in nETS with increasing radius tends to be smaller for GC than for the WRFs, indicating that the skill of GC is less enhanced compared with the skill of the WRFs when positional errors are allowed to some extent. Particularly, for the 70-mm and 100-mm thresholds, the increases in nETS with increasing radius are much smaller for GC than for the WRFs. When the radius increases from 10 to 50 km, the nETSs of GC and WRF_{27km} increase by 0.20 and 0.32 (0.14 and 0.35), respectively, at the 70-mm (100-mm) threshold. The allowance of positional errors leads the WRF with a similar resolution (WRF_{27km}) to outperform GC as well as significantly enlarging the difference in nETS between GC and the WRFs with higher resolutions (WRF_{9km} and WRF_{3km}).

Figure 9 shows the performance diagram by different precipitation thresholds for the GC, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations, which supplements the evaluations conducted above. The TS monotonically decreases with increasing precipitation threshold for all models, indicating decreasing tendencies of prediction skill with increasing precipitation magnitude. WRF_{3km} and WRF_{9km} show the largest and second largest TSs, respectively, among the models at the heavy precipitation thresholds (≥70 mm). The FB of WRF_{3km} is slightly smaller than 1 at most precipitation thresholds, indicating slight underprediction of precipitation of most magnitudes. The FBs of WRF_{9km} and WRF_{27km} are slightly larger than 1 at precipitation thresholds smaller than 30 mm and are smaller than 1 at the heavy precipitation thresholds, indicating slight overprediction of light precipitation and underprediction of heavy precipitation. The FB of GC at precipitation thresholds smaller than 30 mm is even larger than those of WRF_{9km} and WRF_{27km}, indicating more considerable overprediction of light precipitation compared with the WRFs. This is consistent with the wider areas of appreciable precipitation (> 5 mm) in GC than in the observation (Figs. 2 and 3). Meanwhile, GC exhibits significantly smaller FB than the WRFs at the heavy precipitation thresholds, showing lower POD. The FB (POD) averaged from the 70- to 100-mm thresholds is 0.22 (0.16) for GC and is 0.61 (0.26) for WRF_{27km}. The results indicate considerable underprediction of heavy precipitation by GC which agrees with the misses of observed precipitation peaks with scales finer than ~100 km by GC (Figs. 2 and 3) and results in low prediction skill scores for heavy precipitation (Figs. 7 and 8). The overall results from Figs. 7, 8, and 9 suggest that GC has lower skill in predicting heavy/extreme precipitation than in predicting general patterns of precipitation, due to the weak ability to reproduce precipitation peaks.

As model resolution increases, simulated precipitation can capture finer-scale characteristics of observed precipitation. To examine whether the performance differences between the models are attributed to fine-scale characteristics of precipitation fields that cannot be resolved by the resolutions of ~0.25° (or ~25 km), their prediction performances are compared after the resolutions of precipitation data from WRF_{9km} and WRF_{3km} are adjusted to 27 km. This examination can help to better understand the performance difference shown in the comparisons between the GC and the WRFs. When the resolutions are adjusted, the scores for NMB, NCRMS, and SCC are very similar to the results presented in Fig. 6 (Fig. S1). The magnitudes of NMBs in GC are similar to those in WRF_{27km} and slightly larger than those of WRF_{9km} and WRF_{3km} (Fig. S1a). The NCRMSs in GC are smaller than those in the WRFs (Fig. S1b). The SCCs in GC are higher than those in WRF_{27km}, similar to those in WRF_{9km}, and lower than those in WRF_{3km} (Fig. S1c). The

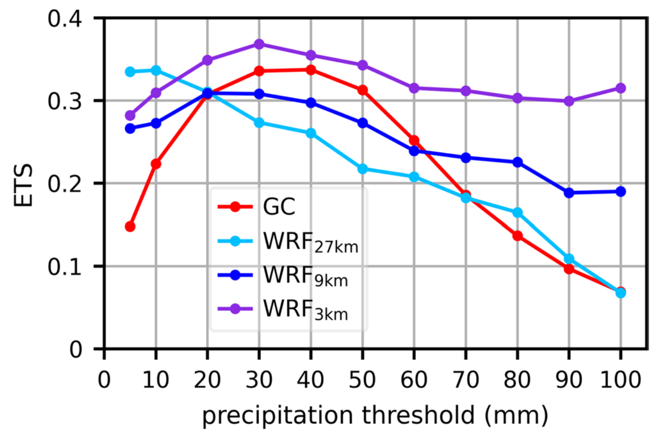

Fig. 7 Equitable threat scores with different precipitation thresholds for the GraphCast, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations

similarity between the results of Figs. 6 and S1 is to some extent expected since these metrics mainly evaluate general patterns of simulated precipitation across South Korea.

Meanwhile, the results for ETS, nETS, and performance diagram are also largely consistent with the results presented in Figs. 7, 8, and 9 although the resolutions are matched (Figs. S2–S4). At the moderate precipitation thresholds (30–60 mm), the ETS of GC is higher than those of WRF_{27km} and WRF_{9km} and lower than that of WRF_{3km} (Fig. S2). At the heavy precipitation thresholds (70–100 mm), the ETS of GC is similar to or slightly lower than that of WRF_{27km} and is appreciably lower than those of WRF_{9km} and WRF_{3km}. At the 70-mm and 100-mm thresholds, the increases in nETS of GC due to the radius increase from 10 to 50 km (0.20 and 0.14, respectively) are noticeably smaller than the increases in nETS of WRF_{27km} (0.32 and 0.35, respectively), WRF_{9km} (0.31 and 0.34, respectively), and WRF_{3km} (0.38 and 0.37, respectively) (Fig. S3). The FB of GC at the heavy precipitation thresholds is much smaller than the FBs of the WRFs at the same thresholds (Fig. S4). The qualitatively consistent results of Figs. 7, 8, and 9 and Figs. S2–S4 indicate that the differences in prediction skill for heavy/extreme precipitation between the models are mainly attributed to their prediction skill for precipitation peaks with scales coarser than ~25 km, which can be represented by the resolutions of precipitation data from GC and WRF_{27km} as well as those from WRF_{9km} and WRF_{3km}.

The weakness of GC is partly reflected in the values of a metric evaluating general patterns of precipitation for individual cases as well. Figure 10 shows the NCRMS of each case for the GC, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations. Here, the cases are arranged from rank 1 to 10, in descending order according to the standard deviation of observed precipitation amounts larger than 70 mm. This standard deviation represents the spatial variability of observed heavy precipitation, which is closely associated

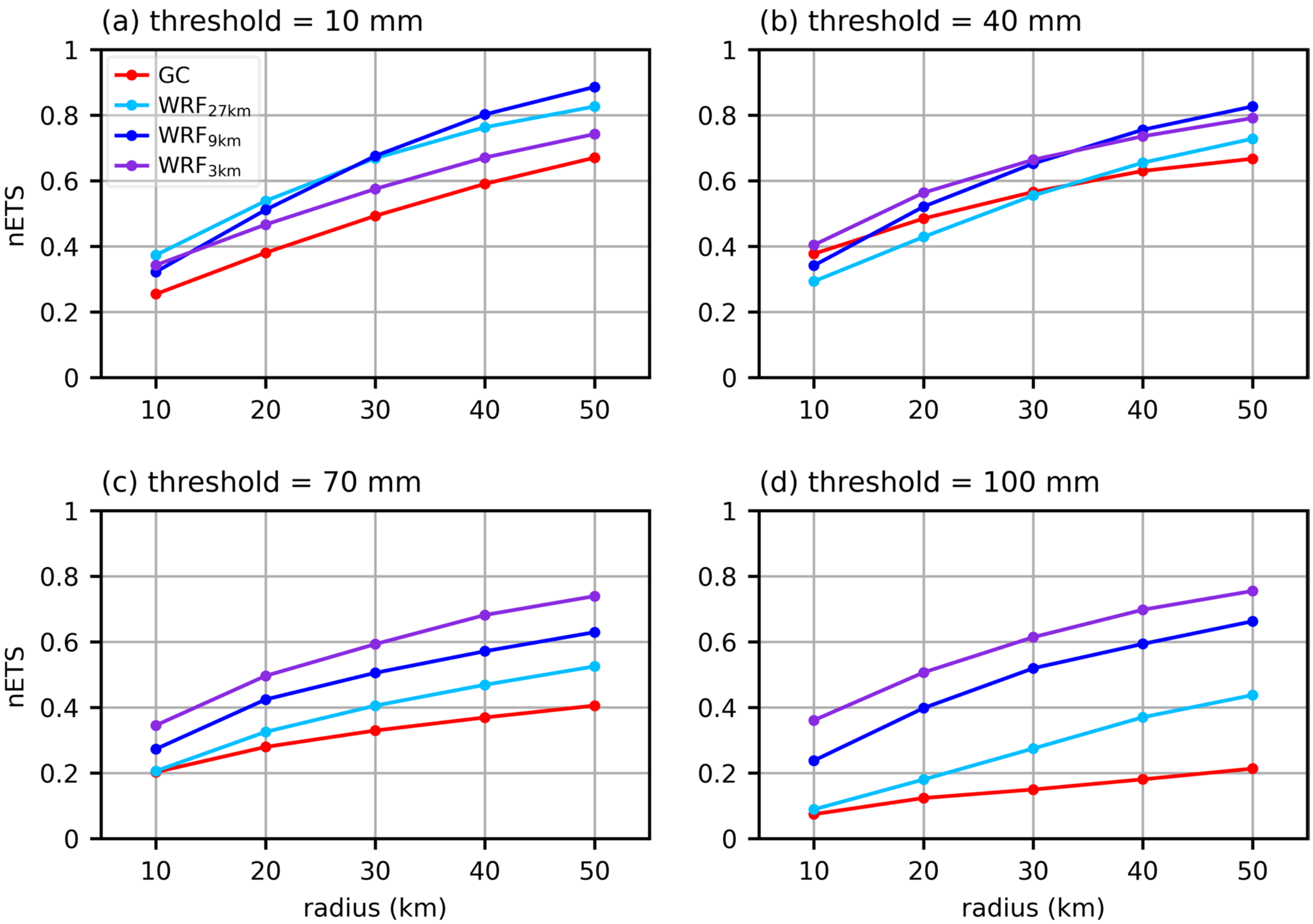

Fig. 8 Neighborhood equitable threat scores as functions of radius with precipitation thresholds of (**a**) 10, (**b**) 40, (**c**) 70, and (**d**) 100 mm for the GraphCast, WRF$_{27km}$, WRF$_{9km}$, and WRF$_{3km}$ simulations

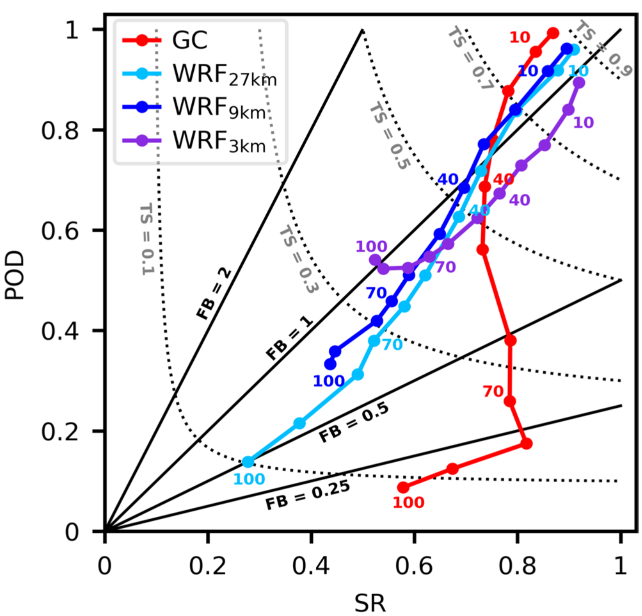

Fig. 9 Performance diagram by different precipitation thresholds for the GraphCast, WRF$_{27km}$, WRF$_{9km}$, and WRF$_{3km}$ simulations

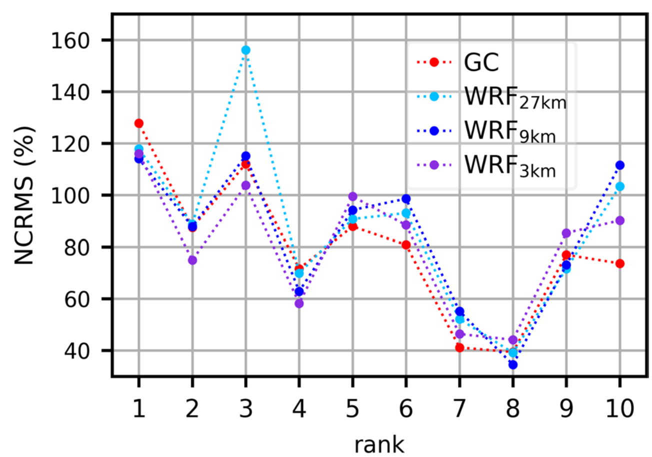

Fig. 10 Normalized centered root-mean-square errors of each heavy precipitation case for the GraphCast, WRF$_{27km}$, WRF$_{9km}$, and WRF$_{3km}$ simulations. The cases are arranged in descending order according to the standard deviation of observed precipitation amounts larger than 70 mm

with precipitation peaks. GC consistently exhibits larger NCRMS than WRF_{3km} for all the 4 cases with the largest standard deviations and shows relatively large or very similar NCRMS compared with WRF_{9km} and WRF_{27km} for 3 out of the 4 cases. This shows that when the heavy precipitation cases having prominent precipitation peaks are selected, GC exhibits worse scores even for a metric evaluating general patterns of precipitation. Meanwhile, GC consistently exhibits smaller NCRMS than WRF_{3km} for all the 6 cases having relatively small standard deviations and shows smaller NCRMS than WRF_{9km} and WRF_{27km} for 4 out of the 6 cases.

The weak ability of GC to reproduce heavy/extreme precipitation in the 10 cases is qualitatively in line with the results from the recent studies of Radford et al. (2025a); Pan et al. (2025) reporting limitations of AIWP models in reproducing intense precipitation (>10 mm day^{-1}) or precipitation variabilities. This weakness could be partly inherited from the training data, that is, the ERA5 data which struggle to capture heavy precipitation (Lavers et al. 2022). Moreover, the use of mean square error as an objective function in the training of AIWP models (Lam et al. 2023; Lang et al. 2024) can also contribute to the underprediction of precipitation maxima, as the use of mean square error can blur variabilities having relatively small scales due to double penalty effect (e.g., Bonavita 2024). The impact of minimizing mean square errors was clearly shown by DeMaria et al. (2025) for predicting the maximum wind speed of tropical cyclone.

3.3 Topographic effect on precipitation

Whether AIWP model produces physically consistent simulation results is a significant issue. Topography is one of the important physical factors influencing precipitation; it can have crucial impacts on local precipitation accumulation in regions with complex terrain for heavy precipitation events. In many of the heavy precipitation cases examined in this study as well, accumulated precipitation amounts near the Sobaek Mountains (dotted box in Fig. 1b) are relatively large compared with those in other regions with similar latitudes in South Korea (Figs. 2 and 3), which is likely due to topographic enhancements of precipitation by the Sobaek Mountains (e.g., Hyun et al. 2010; Jo et al. 2020). In this subsection, how well topographic enhancements of precipitation are reproduced by GC is examined. To examine the topographic enhancements due to the Sobaek Mountains, the regions in 33.8°–36.3°N and within 40 km of the terrain higher than 750 m are named TP (contoured in pink in Fig. 1b) and other regions in 33.8°–36.3°N are named NT (contoured in sky-blue in Fig. 1b). For this, the terrain height data of the WRF_{3km} simulations are used.

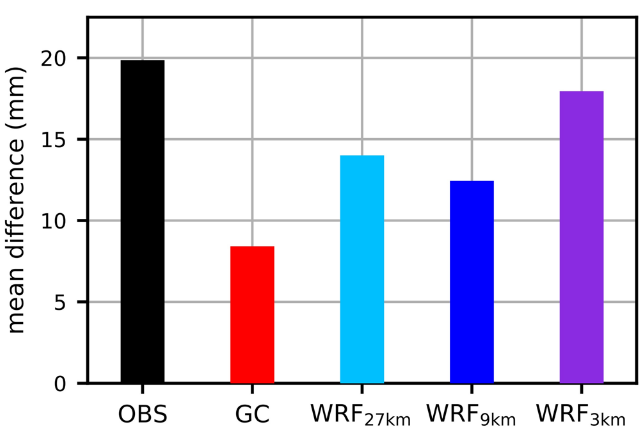

Fig. 11 Differences in 24-h accumulated precipitation amount between the TP region and NT region averaged over Cases 1–10 in the rain gauge observation and in the GraphCast, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations

Figure 11 shows the differences in 24-h accumulated precipitation amounts between the TP and NT regions in the rain gauge observation and in the GC, WRF_{27km}, WRF_{9km}, and WRF_{3km} simulations. In the observation, the TP region receives 20 mm more precipitation than the NT region, on average, in Cases 1–10, which implies topographic enhancements of precipitation. The difference in precipitation amounts between the TP and NT regions simulated by GC is 8 mm, which is 42% of the observed difference. Meanwhile, the precipitation difference simulated by WRF_{27km} is 14 mm (71%), which is larger than the precipitation difference simulated by GC. The precipitation difference for WRF_{9km} is 12 mm (63%) and that for WRF_{3km} reaches 18 mm, which is 90% of the observed difference. These results suggest that GC struggles to reproduce topographic enhancements of precipitation compared with the WRFs in the heavy precipitation cases of interest.

To estimate how much the precipitation field characteristics with scales finer than ~25 km contribute to the performance difference in topographic enhancements of precipitation, the precipitation difference between the TP and NT regions is examined after the resolutions of precipitation data from the WRFs are matched to 27 km (Fig. S5). With the adjusted resolutions, the precipitation simulated by the WRFs reproduces the observed topographic enhancement by 64–88%, which is consistently higher than the result of GC (42%). This indicates that the performance difference in topographic precipitation enhancements between GC and the WRFs is mainly attributed to their prediction skill at ~25-km or coarser scales.

The synoptic condition and its associated water vapor transport averaged over the heavy precipitation cases in the GC and WRF_{27km} simulations are presented in Fig. 12. The prediction results at 0600, 1200, 1800, and 2400 LST on the heavy precipitation days are used for the average over the cases. The sea-level pressure is relatively low west of the Korean Peninsula and is relatively high southeast of the Korean Peninsula, which is associated with low-pressure systems approaching from the west and the western North Pacific subtropical high expanded from the southeast. Southwesterlies developing under this pressure distribution transport a large amount of water vapor, producing atmospheric river that passes over South Korea and exhibits a peak intensity exceeding 800 kg m^{-1} s^{-1} southeast of South Korea. The pressure systems and resultant strong atmospheric river are a typical condition of warm-season heavy precipitation in South Korea (Park et al. 2021). The synoptic conditions and their associated moisture transports simulated by GC and WRF_{27km} are very similar (Fig. 12a and b). The integrated water vapor transport averaged over the red box in Fig. 12 is 698 and 722 kg m^{-1} s^{-1} for GC and WRF_{27km}, respectively, being marginally larger (3%) for WRF_{27km} than for GC. This shows that the intensity of synoptic water vapor transport into the southern region of South Korea for GC and WRF_{27km} is close to each other.

Despite the similar synoptic conditions, the intensity of upward motions above the Sobaek Mountains is differently simulated by GC and WRF_{27km}. Figure 13 shows the fields of pressure vertical velocity and wind vector in the red box in Fig. 12 for Cases 1–10 in the GC and WRF_{27km} simulations. Along the southwesterlies, the moist air passes over the Sobaek Mountains from the west to the east (Figs. 12 and 13). Upward motions stronger than 0.3 Pa s^{-1} are found over the southern region of South Korea in both the simulations. It is noted that the upward motions above the Sobaek Mountains are considerably stronger for WRF_{27km} than for GC, with their intensity well exceeding 0.5 Pa s^{-1} for WRF_{27km}. Although the Sobaek Mountains represented in the two simulations are seemingly different from each other in the pressure coordinate, the differences in their geometric height are small (< 32 m). The considerable intensity difference in upward motions above the Sobaek Mountains in spite of the similar synoptic conditions and mountain heights suggests that the topographic enhancements of upward motions are less predicted by GC than by WRF_{27km}, which is in line with more pronounced underprediction of topographic enhancements of precipitation by GC (Fig. 11). A deficiency in simulating the topographic effects is a type of lacks in physical consistency which requires further investigation in other regions as well. Other lacks in physical consistency of AIWP models were also reported by Bonavita (2024), in terms of the geostrophic wind balance and the rotational and divergent wind components.

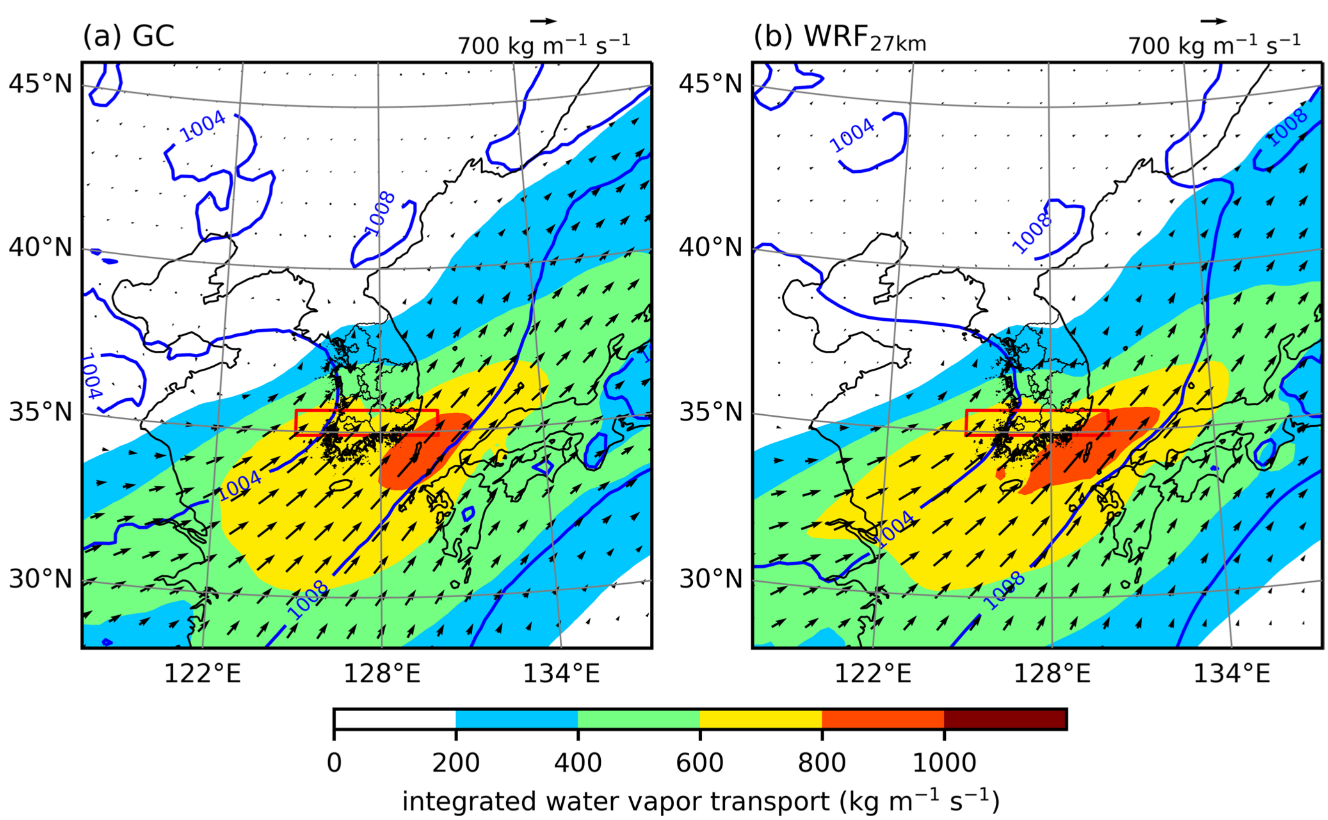

Fig. 12 Fields of integrated water vapor transport (shades and arrows) and sea-level pressure (blue contours) averaged over Cases 1–10 in the (**a**) GraphCast and (**b**) WRF_{27km} simulations

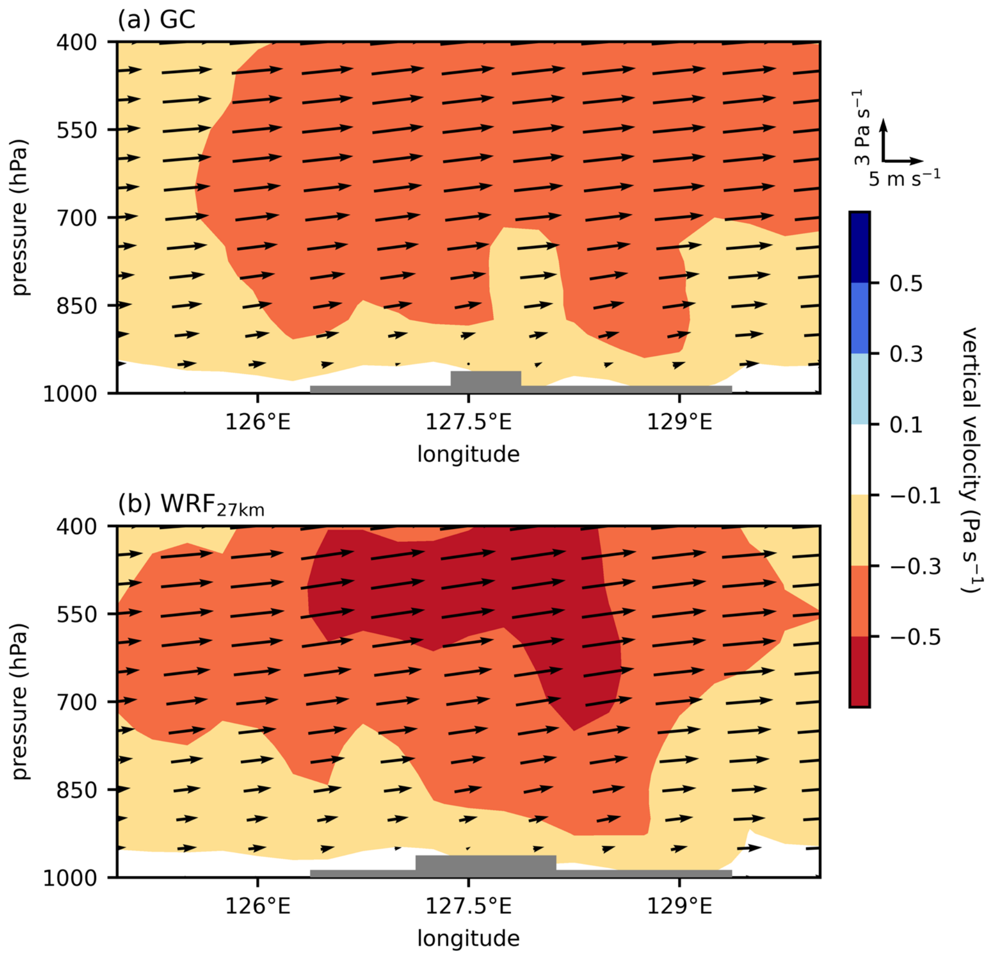

Fig. 13 Fields of pressure vertical velocity (shades) and wind vector (arrows) averaged over the latitudes within the red box in Fig. 12 for Cases 1–10 in the (**a**) GraphCast and (**b**) WRF$_{27km}$ simulations. The gray-shaded areas represent the topography in the red box in Fig. 12

The initialization time of GC (i.e., 0000 LST on the day of interest) is different from that of the WRFs (i.e., 1200 LST on the previous day) in this study. This initialization time of GC is used to examine the best attainable performance of GC with the shortest possible lead time, as GC does not need a spin-up period which is needed for the WRFs. To estimate potential impacts of different initialization times on the results, an additional GC simulation initialized at 1200 LST on the previous day for each heavy precipitation day (GC$_{12LST}$) is conducted. Compared with the WRFs, GC$_{12LST}$ shows larger NMB magnitudes and exhibits comparable or better scores of NCRMS and SCC (Fig. S6). GC$_{12LST}$ shows relatively high (low) ETS at the moderate (heavy) precipitation thresholds (Fig. S7) and exhibits the precipitation difference that is smaller than the precipitation differences for the WRFs (Fig. S8). Overall results from the GC$_{12LST}$ simulation are qualitatively in line with the results from the original GC simulation. It is noted that the ETS at the heavy precipitation thresholds and the precipitation difference between the TP and NT regions are to some extent smaller for GC$_{12LST}$ than for the original GC (Figs. 7, 11, S7, and S8). This is likely associated with the longer lead time of GC$_{12LST}$ compared with that of the original GC.

4 Summary and conclusions

In this study, the prediction skill of GC is examined against the rain gauge observation focusing on heavy precipitation in the summer of 2020 in South Korea, which broke records and resulted in substantial casualties and economic losses. The 10 days exhibiting the largest 24-h accumulated precipitation amounts over South Korea in the summer of 2020 are selected, and for comparisons, the WRF simulations with different resolutions are used. For the NMB, NCRMS, and SCC which mainly evaluate general patterns of simulated precipitation, GC exhibits higher or competitive scores compared with WRF$_{27km}$, WRF$_{9km}$, and WRF$_{3km}$. GC also well predicts synoptic conditions and their associated moisture transports in the cases. Meanwhile, the ETS, nETS, and other contingency table-based metrics suggest that GC has lower skill in predicting heavy/extreme precipitation than WRF$_{27km}$ as well as WRF$_{9km}$ and WRF$_{3km}$. The precipitation patterns in the 10 cases and supplementary analysis results with resolution adjustment imply that the relatively low skill scores of GC for heavy/extreme precipitation in these 10 cases are associated with the weak ability to reproduce precipitation peaks with scales of ~25–100 km that are associated with large local accumulations. The precipitation enhancements associated with the Sobaek Mountains are reproduced up to 42% by GC which is lower than those reproduced by the WRFs (63–90%) for the 10 cases. Although the synoptic conditions and their associated water vapor transports simulated by GC and WRF_{27km} are similar to each other, the upward motions over the Sobaek Mountains simulated by GC are weaker than those simulated by WRF_{27km}.

The overall results of this study show that GC well predicts the general patterns of precipitation and synoptic conditions in heavy/extreme precipitation events. Given the enormously short computation times of GC prediction compared with the prediction from NWP model, this indicates great potential of AIWP model for forecasts of heavy/extreme precipitation events. Meanwhile, the results of this study also imply that GC may underestimate heavy/extreme precipitation peaks and topographically enhanced precipitation, both of which can be viewed as mesoscale variabilities of precipitation. This is qualitatively in line with the reported difficulties of AIWP models in reproducing mesoscale temperature and wind variabilities (Bonavita 2024; Charlton-Perez et al. 2024; DeMaria et al. 2025). As the mesoscale maxima (e.g., precipitation peaks, maximum wind speeds) are likely to be associated with the severest hazards in extreme weather events, the prediction of mesoscale variabilities is important as well. Investigations of proper objective functions as well as constructions of reliable open-access high-resolution training datasets (e.g., datasets from global convection-permitting model) could be potentially beneficial for further improving AIWP skill in heavy/extreme precipitation peaks. Considering strengths and weaknesses of AIWP models, Radford et al. (2025a) suggested a forecast strategy that AIWP model is used to forecast the most likely locations of precipitation features and NWP model is used to forecast intense precipitation within the features. Also, there are recent studies that attempted to combine AIWP and NWP by developing a hybrid model that uses governing equations for dynamics and machine learning for physics parameterizations (Kochkov et al. 2024) or by driving high-resolution NWP model using AIWP results (Xu et al. 2024). Further investigation of hybrid forecast strategies incorporating the merits of AIWP and NWP would be helpful for introducing AIWP into operational forecasting.

In this study, ensemble WRF simulations are chosen as a counterpart of a single GC simulation for each heavy precipitation event. This choice is made, as WRF simulation exhibited a high sensitivity to parameterization options and small initial condition perturbations even for short lead times while GC simulation did not show a high sensitivity to small initial condition perturbations for short lead times. This characteristic of GC led ensemble approach using small initial condition perturbations to be ineffective for 1-day GC simulation which is of interest in this study (see Figs. S9–S12). Nevertheless, the inconsistency in the comparisons might hinder a more detailed examination of AIWP skill, being a limitation of this study. An examination based on more consistent comparisons and further investigation of ensemble effects on AIWP are needed. This study looks into the precipitation prediction of GC in the heavy precipitation cases occurred in 2020. While the examination can contribute to the understanding of AIWP skill in heavy/extreme precipitation events, the prediction skill identified from these events cannot represent the overall skill in predicting a wide range of precipitation events. The general performance of AIWP model in predicting precipitation over East Asia deserves further investigation. The analysis results for topographic enhancements of precipitation in this study are obtained from an examination of 10 precipitation cases in South Korea. A more systematic assessment using a larger number of cases with their categorization according to the wind type for mountains in various regions will be beneficial for more robust examinations of the performance of GC in simulating topographically enhanced precipitation.

This study uses the WRF model, which is different from many studies using the IFS model (e.g., Radford et al. 2025a; Pan et al. 2025), for conducting numerical weather simulations with different resolutions. This enables the comparisons of AI-based simulations with numerical simulations having different degrees of prediction skill in heavy precipitation. Meanwhile, GC and the WRF model have differences in many aspects, beyond their AI- or physics-based approaches. The training dataset of GC is produced by the IFS model employing hydrostatic equations while the WRF model employs nonhydrostatic equations. Also, simulations from the WRF model are affected not only by initial conditions but also by boundary conditions from the parent domain, unlike those from the IFS model. These differences are closely associated with the model capability for higher-resolution (< 10 km) simulations, thereby being difficult to be avoided in this study. Hence, in this study, acknowledging the differences, simulations are conducted using the same dataset (i.e., ERA5 data) as initial and/or boundary conditions. Given the multifaceted differences between the two models, the results of this study should not be interpreted as indicating either superiority or inferiority of AI-based approach over physics-based approach in general but as an examination of the skill level that current AIWP model has reached through comparisons with numerical weather simulations having various degrees of skill.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00704-026-06536-w.

Acknowledgements We greatly appreciate two anonymous reviewers providing valuable comments on this study. We thank the research group of GraphCast for openly providing the source codes of GraphCast. We are also grateful to the ECMWF for publicly distributing libraries to use AI-based weather prediction models including GraphCast as well as the ERA5 data.

Authors' contributions Jong-Jin Baik designed this study. Seong-Ho Hong performed the data analysis and visualization. All authors discussed the results. Seong-Ho Hong wrote the original manuscript. Kyeongjoo Park, Dong-Hwi Kim, Jong-Jin Baik, and Han-Gyul Jin reviewed and edited the manuscript. All authors read and approved the final version of the manuscript.

Funding This work was funded by the National Research Foundation of Korea (NRF) under grant RS-2025-00562044. Han-Gyul Jin was supported by a New Faculty Research Grant of Pusan National University, 2023, and by the Global–Learning & Academic research institution for Master's·PhD students, and Postdocs (LAMP) Program of the NRF grant funded by the Ministry of Education (No. RS-2023-00301938).

Data availability The codes of GraphCast model used in this study are downloaded from GitHub (https://github.com/google-deepmind/graphcast). The ERA5 data are available in the Copernicus Climate Change Service (https://cds.climate.copernicus.eu), and the rain gauge data are available on the KMA Weather Data Service (https://data.kma.go.kr). The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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